

Trident G: A Computational-Level Theory of Intelligent Life

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December, 2025

Abstract

Trident G is a general theory of intelligent life that reframes brains and other adaptive systems as regulators of a viable operating band, rather than as pure surprise-minimising machines. Instead of driving free energy F towards zero, an agent defends a non-zero allostatic set-point $F^\star$ within a viability band where it can think, learn and act without collapse. Around this defended budget, we define a near-critical Ψ -band over a cusp landscape with state axis $x = F - F^\star$ and control axes for expected challenge ξ and explore–exploit tilt δ . General intelligence is then the navigable $\mathcal{G}$ -volume of this Trident-shaped space that an organism can reach, traverse and re-enter while remaining within its Ψ -band. I outline how this template can be instantiated by different physical substrates, how meta-variables for efficacy, coherence and niche coupling modulate Ψ -occupancy across time, and how the framework generates concrete predictions for resilience, ageing and training. Trident G thus treats intelligence as life on the cusp: flexible regulation of a near-critical band in a changing, challenging world.

1. The Overarching Computational Problem

Any viable organism faces two top-level requirements:

1. Stay alive: keep its overall free-energy / allostatic load within a viability set where the body does not fail.
2. Exploit and learn from its niche: not merely avoid death, but adapt, explore, and improve its models and skills over time.

Trident G assumes that organisms do not simply aim for a single fixed set-point, but for a *viability band* of states in which both survival and flexible intelligence are supported. In other words, the system's objective is to occupy a corridor of good functioning rather than to drive its free-energy / load to a minimum, because a certain amount of challenge and uncertainty is required for learning, plasticity and long-term fitness.

In this paper, free energy is used in the active-inference sense: a variational free-energy functional that upper-bounds surprise under the organism's generative model. High free energy corresponds to large prediction errors and costly corrective action; low free energy corresponds to a well-regulated, energetically efficient state. At the organism level, this variational free energy is understood to be tightly coupled to *allostatic load* – the metabolic and regulatory cost of keeping key variables within bounds. For simplicity, Trident G therefore treats F as a single scalar summarising an overall mind–body budget: an allostatic / free-energy load that combines the cost of maintaining physiological stability and the cognitive work needed to keep prediction errors manageable. In the interoception literature this is sometimes called a “body budget”, but here we emphasise that cognitive and epistemic costs are part of the same regulated budget.

1.1 Viability Set, Griffiths Zone and Ψ -Band

We distinguish three nested sets:

Viability Set

A broad interval of allostatic load:

$$F_{min} \leq F \leq F_{max}$$

within which the organism remains alive over the relevant horizon (hours–years). Outside this set – too little or too much load – you get death or irreversible damage. Here F should be read as overall allostatic / free-energy load at the organism level, integrating metabolic, physiological and (where present) cognitive contributions to this mind–body budget.

Griffiths Zone / Viability Band Around F ★

Within the viability set there is a narrower near-critical band where the underlying regulatory dynamics live in an extended Griffiths-like phase: high dynamic range, rich variability and good responsiveness. This is where the system can flexibly reconfigure without falling into either rigid order or chaos. At the substrate level this corresponds, in neural organisms, to regimes where neural, autonomic and endocrine networks exhibit long correlation lengths and power-law-like variability, and more generally to regimes where the organism’s regulatory networks support rapid reorganisation without loss of coherence.

Ψ -Band

Within the Griffiths zone there is a still narrower operational corridor where:

- net utility

$$U(F) = B(F) - C(F)$$

(benefit minus cost) is high,

- recovery from perturbations is reliable, and near-critical dynamics support fluid reasoning, controlled action and creative reconfiguration (for organisms with such capacities).

This Ψ -band is what the control system actively tries to occupy on behavioural time-scales: not just *alive vs dead*, but *sustainably near-critical and capable*. In later sections, the cusp dynamics and meta-parameters are defined precisely to bias the system towards this Ψ -band rather than merely hugging the centre of the broader viability set.

1.2 The Allostatic Objective: Band, Not Point

At the computational level we summarise the organism's outer objective by a simple loss:

$$\mathcal{L} = 1/2 (F - F^\star)^2$$

This should be read as a shorthand for: Keep actual allostatic load F within a defended band around a near-critical reference $F^\star$, rather than minimising load to zero.

Formally, this quadratic treats $F^\star$ as a central reference, but in Trident G $F^\star$ parameterises the centre of a band covering the Ψ -region, not a single exact physical set-point. The quadratic loss is a convenient local approximation to a flatter-topped utility landscape where states across the Ψ -band all have high value.

There are two important implications:

- When $F \ll F^\star$ (over-safe, under-stimulated, rigid so-called “dark-room” regimes), the organism should seek additional uncertainty and complexity.
- When $F \gg F^\star$ (chronic overload), it should reduce uncertainty and complexity.

In other words: maintaining near-criticality in a band, rather than minimising deviation, is the core ‘life on the cusp’ idea. Later, this band-based objective is implemented using cusp

dynamics whose control parameters steer the system towards a near-critical Ψ -region on a richer F – G landscape.

2. Core Quantities

The theory is built around four primary quantities:

$F(t)$: Actual Free Energy / Allostatic Load

The current mind-body budget cost the organism is paying, aggregating metabolic, physiological and (where relevant) cognitive strain. You can think of this as the realised free-energy / allostatic burden at time t , expressed as a single scalar.

$G(t)$: Expected or Predicted Load

How much load the system is prepared for at this moment, given its current beliefs about what the task/niche will demand. This is the anticipatory/allostatic part. In Free Energy Principle terms, $G(t)$ can be read as an expected free energy–like quantity: a prediction of upcoming allostatic cost under the organism’s current model and policies, expressed on the same scale as $F(t)$. These expectations are informed by both signals about the external niche or task (exteroceptive channels) and signals about the internal milieu (interoceptive channels), in whatever form those channels take for that organism.

$F^\star(t)$: Defended Allostatic Set-Point

The centre of the viability band – a reference value for *healthy enough, near-critical operation* given the organism’s current traits, health and niche. This encodes where the organism would like its actual load to sit when things are going well, i.e. the centre of its Ψ -band corridor.

$G^\star(t)$: Baseline Expected Load

The typical, slowly learned expectation of how demanding this task/niche usually is (e.g. ‘this environment is normally high-demand’, ‘this task is usually easy’).

Together, $F^\star$ and $G^\star$ define the long-run expectations for where the organism wants to sit and how demanding it thinks life normally is. They are slow variables, shaped by traits, history and context, against which the faster F and G signals are evaluated.

We assume a single generative model that jointly represents hidden causes of both external (exteroceptive) and internal (interoceptive / affective) signals. In Trident G, the quantities F , G , $F^\star$ and $G^\star$ are coarse summaries of this joint model's beliefs about current and expected allostatic load, and emotion corresponds to valenced salience over these internal causes as they relate to the defended Ψ -band.

By inference we mean the ongoing process whereby the organism updates these beliefs about hidden causes and about the consequences of candidate actions so as to keep F and G aligned with its defended viability band. In simple organisms this may amount to adjusting a small number of reflex gains or set-points; in human and other cognitively sophisticated organisms the same machinery is elaborated into multi-step relational inference and explicit reasoning, allowing the system to simulate counterfactual trajectories in both exploratory and exploitative modes while still serving the same underlying allostatic objective.

From these we construct the derived signals.

State deviation:

$$x(t) = F(t) - F^\star(t)$$

How far am I from my defended band right now?

This is a pure state measure: positive x means *currently overloaded relative to my defended band*; negative x means *currently under-stimulated / under-critical*.

Total engagement / expected challenge:

$$\xi(t) = G(t) - F^\star(t)$$

How much load am I prepared to carry, relative to the same band?

This is not *how loaded I am*, but *how loaded I'm willing or expecting to be*, measured against the same reference $F^\star$. It is the stance variable: risk appetite / challenge endorsement in that moment.

It is often useful to split ξ into a slow baseline and a fast adjustment:

Tonic engagement bias:

$$\zeta^\star(t) = G^\star(t) - F^\star(t)$$

Long-run challenge stance in that niche: are you generally cautious, or do you habitually run *hot* and seek high challenge?

Phasic engagement update:

$$\check{\xi}(t) = G(t) - G^\star(t)$$

Short-term adjustment around that stance: *for this episode, I'll push a bit harder or for now, I'm backing off.*

So, by construction:

$$\xi(t) = \zeta^\star(t) + \check{\xi}(t).$$

This lets the theory distinguish trait-like challenge preferences ($\xi^\star$) from state-like momentary mobilisation ($\check{\xi}$) while keeping a single effective engagement term ξ for the cusp dynamics. In later sections, ξ_{eff} is constructed from these components to determine whether the cusp opens into multiple modes or collapses into a single basin.

Allostatic prediction error

$$\varepsilon(t) = F(t) - G(t) = x(t) - \xi(t)$$

How wrong was my expectation about load?

- If $\varepsilon > 0$, the niche is harsher than expected (you are under-braced).

- If $\varepsilon < 0$, the niche is easier than expected (you are over-braced).
- If $\varepsilon \approx 0$, reality broadly matches your endorsed challenge level.

Intuitively:

- $F - F^\star$ (or x) encodes the state: where your actual mind-body budget sits relative to the defended band.
- $G - F^\star$ (or ξ) encodes your stance: how much challenge you currently endorse or expect, relative to that same band.
- Their difference $\varepsilon = x - \xi$ is the allostatic prediction error that drives:
 - learning (updating $G^\star, F^\star$),
 - reconfiguration of mode structure on the cusp (via ξ_{eff}, δ), and
 - shifts in engagement ($\tilde{\xi}$) on shorter time-scales.

This same ε signal also feeds into valuation and salience systems (e.g. via valence tags and σ -based fields). At a generic level, a salience–allostasis hub pools prediction errors from:

- internal channels that track the organism’s own bodily state (signals about the internal milieu), and
- external channels that track the niche (signals from the sensory surface and current perception–action context).

It then writes σ -fields over both world-model / task representations and perception–action policies, determining whether a given deviation from expectation is treated as enriching exploration or harmful overload, and helping to set the value difference $\Delta V(t)$ that shapes the explore–exploit tilt δ . (In humans, this functional division roughly maps onto interoceptive and exteroceptive networks feeding into large-scale control systems such as FPCN-A, which is more

coupled to default-mode / semantic maps, and FPCN-B, which is more coupled to dorsal attention and sensorimotor hierarchies.)

At the implementation level, we assume that a valence-coded prediction signal derived from $\varepsilon(t)$ is carried, in mammals, largely by phasic dopamine. When outcomes are better than expected in allostatic and epistemic terms (lower realised cost, or higher reward / model fit than anticipated), midbrain dopamine bursts bias plasticity towards D1-dominant *Go* channels and away from D2-dominant *NoGo* channels, consolidating the currently successful policy. When outcomes are worse than expected, dopamine dips have the opposite effect, favouring D2-mediated plasticity, weakening the just-used policy and making alternatives more competitive. In Trident G we treat this opponent D1–D2 system as a neural realisation of the explore–exploit tilt δ : D1-dominant states support Executive, exploit-based control; D2-dominant states open the door to Creative, exploratory search over alternative models and strategies.

3. Marr’s Computational Level Justification

At Marr’s computational level, the organism’s problem is:

- Stay in a viability band: keep F near a near-critical defended band $F^\star$, not at zero;
- Decide how far from that band it is willing to roam for epistemic gain, reward, training, and so on.

That strongly implies at least two latent degrees of freedom:

Current deviation:

$$x = F - F^\star$$

Where am I relative to my defended band?

Endorsed or expected deviation:

$$\xi = G - F^\star$$

How much deviation do I expect and accept in this context?

Here F is the realised allostatic or free-energy load, and G is the expected or *allostatically braced for* load (an expected free energy–like quantity, expressed on the same scale). Both are evaluated against the same reference $F^\star$, which parameterises the centre of the defended band.

Consider three hypothetical organisms:

1. Tracks only $F - F^\star$ (pure homeostat).

It can detect deviation and push back, but:

- It cannot distinguish harmful overload from beneficial training stress;
- It cannot represent *I choose to tolerate this stress now because it improves my future position.*

It degenerates towards dark-room solutions or boring set-point maintenance. The best it can do is minimise deviation, without ever representing deliberate excursions for learning, exploration or growth.

2. Tracks only $G - F^\star$ (pure goal or expectation).

This system has a preferred load but no link to reality. It is clearly insufficient for control.

It can write fantasies about desirable stress levels but has no term that tracks the actual mind-body budget and thus cannot correct itself.

3. Tracks both, plus prediction error ε

Now it can:

- detect miscalibration (over-braced versus under-braced);
- treat the same deviation $F - F^\star$ differently depending on stance ξ :
 - under-critical plus high ξ : deliberate challenge or exploration;

- over-critical plus low ξ : abort and retreat.

This is directly analogous to what reinforcement learning and active inference already do:

- a target or value (here G , anchored to $F^\star$);
- a current state (here F relative to $F^\star$);
- a prediction error ε driving adaptation.

So if you accept allostasis and prediction-error learning as generic principles, something functionally equivalent to $F - F^\star$ and $G - F^\star$ falls out from the computational problem.

Trident G simply makes these explicit, then builds the cusp dynamics and meta-parameters around the interplay of state x , stance ξ and error ε .

4. Why a Band, Not a Point

Trident G refines the band intuition by distinguishing:

Viability set $[F_{\min}, F_{\max}]$:

$$[F_{\min}, F_{\max}]$$

Coarse survival constraints:

- If F is too low for too long (extreme under-arousal or metabolic collapse), basic processes fail.
- If F is too high for too long (chronic allostatic overload), you incur cumulative damage.

This outer set is defined by *hard* physiological limits on what organs, tissues and regulatory systems can tolerate over relevant time horizons.

Defended band around $F^\star$

Within $[F_{\min}, F_{\max}]$, evolution and learning shape a preferred operating region where the trade-off

$$U(F) = B(F) - C(F)$$

(benefit versus cost) is near-maximal.

Because both benefit $B(F)$ and cost $C(F)$ are smooth and noisy, $U(F)$ typically has a flattish top rather than a sharp maximum: a range of F where net utility is *good enough*. The centre of that plateau is $F^\star$; the plateau width is the defended band. Control laws that look quadratic in $F - F^\star$ are therefore best read as local approximations around the centre of this plateau, not as literal drives to a single point.

Ψ -band

A narrower subset of that defended band where:

- near-critical dynamics in the organism's regulatory and control networks are actually realised;
- performance on whatever *intelligent* functions the organism possesses (for example, flexible decision-making, problem solving, efficient foraging) is near-optimal;
- recoverability from perturbations is high.

This is the region where the system is not just *healthy enough* in an energetic sense, but actually operating in the cusp regime that supports multiple flexible modes (Autopilot, Executive and Creative for cognitively sophisticated organisms) and rapid, reversible reconfiguration between them.

What sets the width of these bands?

- **Outer viability set** [$F_{\min}$, $F_{\max}$]: This is determined by basic physiology (what ranges of load organs and tissues can tolerate).
- **Defended band:** This is shaped by
 - the curvature of cost and benefit curves;
 - evolutionary safety margins;

- individual traits (fitness and conditioning) that alter how quickly costs escalate;
- long-run learning about typical niche demands (through $F^\star$ and $G^\star$).
- **Ψ -band:** This is constrained by:
 - requirements for near-critical, flexible cognition and control;
 - the organism's meta-states η (meta-efficacy), χ (coherence) and λ (niche coupling), which determine where near-critical operation is sustainable and recoverable;
 - the salience–allostasis machinery that sets precisions and σ -fields, effectively deciding which parts of the defended band get stabilised as workable operating points, given both internal (interoceptive) and external (exteroceptive) conditions.

Training, illness and environment can thus move and stretch the Ψ -band: successful training and robust niches widen it; chronic stress and impoverished niches narrow it. In Trident G, these shifts appear concretely as changes in meta-parameters and in the shape of the cusp landscape, altering how much of the defended band is actually reachable and stable as a high-functioning near-critical region.

5. From Band Control to Cusp Geometry

The allostatic control problem is embedded in a cusp potential over the state axis x :

$$V(x; \xi^\star, \tilde{\xi}, \eta, \chi, \lambda) = b(\xi^\star, \eta) x^4 - a(\xi^\star, \chi) \xi_{\text{eff}}(t) x^2 - \delta(t) x$$

The minus sign on the quadratic term is chosen so that increasing ξ_{eff} deepens high-engagement wells and widens the bistable region, matching the intended *more engagement* $\rightarrow$ *stronger Gf basins* interpretation.

with:

Effective engagement:

$$\xi_{\text{eff}}(t) = \lambda(t)[\xi^{\star}(t) + \tilde{\xi}(t)]$$

(how much of your endorsed challenge actually *gets through* to the niche, once coupling to the environment is taken into account).

Quadratic coefficient:

$$a(\xi^{\star}, \chi) = a_0[1 + \kappa \chi(t)] \exp(\alpha \max(0, \xi^{\star}(t)))$$

which controls how sharply engagement splits Autopilot versus Gf regimes, and how quickly the system enters a bistable region as ξ_{eff} increases.

Quartic coefficient:

$$b(\xi^{\star}, \eta) = b_0 / \eta(t) \exp(\beta \max(0, -\xi^{\star}(t)))$$

which controls how deep and stiff the wells are. Higher meta-efficacy η effectively flattens the potential (shallower wells and more freedom to move); lower η makes the wells deeper and stiffer for a given $\xi^{\star}$.

Explore–exploit tilt:

The explore–exploit tilt $\delta(t)$ is a bounded, meta-gated signal that biases the system towards either the exploration-dominated or exploitation-dominated Gf branch.

In the base formulation this is driven directly by a valence-weighted prediction error:

$$\delta(t) = \delta_1 \chi(t) \lambda(t) \tanh(\gamma \delta v(t) \varepsilon(t))$$

where $v(t) \in \{-1, 0, +1\}$ tags the sign of the outcome and $\varepsilon(t) = F(t) - G(t)$ is the allostatic prediction error. Positively valenced, better-than-expected outcomes ($v\varepsilon > 0$) push δ towards exploitation; negatively valenced, worse-than-expected outcomes ($v\varepsilon < 0$) push it towards exploration.

In the more σ -based formulation, δ is driven by an allostatic value difference between exploitation and exploration. Under the current mixture of models (wk), the salience–allostasis hub estimates the expected σ -value of continuing to exploit versus switching to explore, and computes

$$V_{\text{exploit}}(t), V_{\text{explore}}(t), \Delta V(t) = V_{\text{exploit}}(t) - V_{\text{explore}}(t).$$

Here V_{exploit} and V_{explore} are expectations of the σ -fields over internal task / world models and perception–action policies (roughly, FPCN-A-like maps and FPCN-B-like control loops). Saying that exploitation (or exploration) is *better for allostasis* means that, given the current generative models and drives, that policy family is expected to keep future free energy F closer to the defended band F^* over the relevant horizon. Exploitation typically does this by tightening control and pulling trajectories back towards states where $F \approx F^*$, whereas exploration does it by tolerating temporary excursions away from current expectations in order to discover or refine policies that will keep F closer to F^* in future.

$$\delta(t) = \delta_1 \chi(t) \lambda(t) \tanh(\gamma \delta \Delta V(t))$$

The σ -based tilt is then given by the expression above. The bounded nonlinearity $\tanh(\cdot)$ ensures that even large value differences only produce a finite tilt, while the meta-states $\chi(t)$ (coherence) and $\lambda(t)$ (niche coupling) gate how strongly these differences can reshape the control landscape. If $\Delta V(t) > 0$, exploitation is currently better for allostasis and the landscape tilts towards exploit; if $\Delta V(t) < 0$, exploratory policies are expected to do better and the landscape tilts towards explore. The prediction-error-based base engine is recovered as the special case $\Delta V(t) \propto v(t)\varepsilon(t)$.

5.1 Shallow vs Steep Folds: Stiffness and Splitting

In canonical catastrophe theory, the cusp potential

$$V(x; a, b) = 1/4 x^4 + 1/2 a x^2 + b x$$

has:

- a bifurcation or splitting parameter a , which controls when two wells appear and how wide the bistable region is;
- an asymmetry or bias parameter b , which tilts one well relative to the other.

If you reintroduce a general quartic coefficient $c > 0$,

$$V(x) = c x^4 + 1/2 a x^2 + b x$$

then:

- When c is larger, the wells become steeper, deeper and narrower, producing a sharper, cliff-like fold.
- When c is smaller, the potential flattens and the wells become shallower and broader.

In Trident G terms:

- The stiffness parameter corresponds to $b(\cdot)$ (or, equivalently, an overall gain factor on the potential V);
- The splitting parameter corresponds to the product $a(\cdot) \xi_{\text{eff}}$.

Meta-parameters tune these:

- λ (niche coupling) amplifies the splitting: high λ makes environmental demands cut deeper grooves for a given $G - F^\star$ by increasing ξ_{eff} ,
- χ (coherence) is a natural candidate to scale stiffness $b(\cdot)$ or an overall gain on V : high χ gives steeper, deeper wells (stronger, more integrated modes); low χ flattens the landscape and makes transitions easier,

- η (meta-efficacy) modulates how deep high-capacity wells are allowed to become before costs trigger retreat, via its effect on $b(\cdot)$ and on how aggressively the system stabilises high-engagement Gf basins.

So *gentle and broad* versus *tight and cliff-like* folds correspond to different settings of these stiffness and splitting parameters, mediated by η , χ and λ . These in turn reflect long-run learning about controllability, coherence and coupling to the niche, which is why training, trauma and context can visibly reshape an individual's mode structure on the cusp.

6. Regimes in the $F - F^\star$, $G - F^\star$ Plane

It is helpful to think in the two-dimensional plane (see Figure 1):

- horizontal axis: $x = F - F^\star$ (actual load);
- vertical axis: $\zeta = G - F^\star$ (expected or endorsed load).

The diagonal $x = \zeta$ is zero prediction error ($\varepsilon = 0$): reality is as tough as anticipated.

Different quadrants then correspond to different combinations of *how it actually is* versus *how tough I am braced for it to be*.

6.1 Both Near Zero: Ψ -Band or Good Regulation

In the well-regulated regime, both actual and expected load sit close to the defended band and prediction errors are small.

- $F \approx F^\star$: near the defended band.
- $G \approx F^\star$: not braced for extreme ease or extreme challenge.
- $|x|$, $|\zeta|$, $|\varepsilon|$ all small.

Here:

- $\mathcal{L} = 1/2(F - F^\star)^2$ is low;
- the system operates in or near the Ψ -band:

- flexible switching between Autopilot and Executive or Creative modes (where these exist);
- high capacity at acceptable cost.

In cusp terms, ξ_{eff} and δ sit in ranges that keep the landscape multi-stable but not excessively stiff. The system can lean into Executive or Creative Gf when needed, then relax back into Autopilot without large allostatic swings.

6.2 Over-Braced (Pessimistic or Hypervigilant)

In the over-braced regime, the agent prepares for high load even though its actual mind–body budget remains only moderately taxed.

- **Expect high load:** $\xi \gg 0$.
→ The system is braced for threat, effort or difficulty, as if something demanding is imminent.
- **Actual load moderate:** x smaller.
→ In reality, the current allostatic strain is modest; the environment is not as harsh as the agent is treating it.
- **Reality is easier than expected:** $\varepsilon = x - \xi < 0$.
→ Each episode confirms that things are safer or easier than predicted, generating negative prediction errors.

Consequences:

- **Unnecessary mobilisation (raised arousal and tension).**
→ Resources are chronically “on alert”, burning energy on false alarms.
- **Chronic cost if maintained.**

→ Sustained over-bracing narrows the Ψ -band and increases wear-and-tear on regulatory systems.

- **G is gradually pulled down over learning, reducing ζ (you stop over-preparing).**

→ With repeated evidence that outcomes are better than expected, the generative model can soften its priors, lowering endorsed challenge and gradually relaxing hypervigilant settings.

In the short term, negative ε with positive valence relaxes unnecessary avoidance: *it is not as bad as I thought; I can probabilistically risk more*. At the dopaminergic level this corresponds to consolidating the just-used, unexpectedly successful policy via D1-biased plasticity; behaviourally, relative to the previous hypervigilant stance, the system can now afford to sample a broader set of options around that policy. In Trident G we describe this as δ being nudged away from rigid avoidance and permitting more exploratory episodes around an otherwise stabilised Executive strategy.

Over time this can increase meta-efficacy η : discovering that you can cope. In Trident G, that shows up as shallower, more navigable wells in the cusp and a wider Ψ -band for the same nominal niche.

6.3 Under-Braced (Over-Optimistic or blindsided)

In the under-braced regime, the agent is carrying a high actual load while still expecting life to be relatively easy.

- **Actual load high:** $x \gg 0$.

→ The current mind–body budget is heavily taxed; the system is already operating at a high allostatic cost.

- **Expected load low:** ζ small.

→ The generative model is still tuned to low-demand conditions and has not braced for sustained challenge or threat.

- **Things are worse than expected:** $\varepsilon = x - \xi > 0$.

→ Each encounter brings an unpleasant surprise: the niche is harsher than the model predicts.

Consequences:

- **Positive prediction error drives sharply increased engagement and salience.**

→ Large $|\varepsilon|$ pushes up fast engagement ξ and raises salience weights $s_i(t)$: the system is forced to pay attention and mobilise resources, flagging the current context as urgent and high-stakes. Although the sign of ε is positive, the associated valence tag is typically negative ($v < 0$): these are *costly* surprises.

- **In the immediate term, δ tilts towards tightly exploitative damage-control policies.**

→ Given the current policy ensemble, the salience–allostasis hub typically assigns higher short-horizon allostatic value to policies that tighten control, reduce uncertainty and pull F back towards F^* than to risky exploration. Formally, this means $\Delta V(t) = V_{\text{exploit}}(t) - V_{\text{explore}}(t) > 0$, so $\delta(t) = \delta_1 \chi(t) \lambda(t) \tanh(\gamma \delta \Delta V(t))$ tilts the cusp towards the exploitation-dominated Gf branch. Behaviourally, the system narrows its focus, suppresses exploratory forays and tries to stabilise the situation using whatever strategy has worked best so far.

- **At the same time, negative-valence prediction errors drive D2-biased plasticity.**

→ Because outcomes are worse than expected, dopamine dips favour D2-mediated plasticity in cortico-striatal loops. The currently over-used, failing policy is progressively weakened and competing policies gain relative strength. In value terms, repeated

overload episodes drive its σ -based evaluations downward, reducing $V_{exploit}$ for that strategy family.

- **Over learning, the stance and landscape are reshaped.**

→ As the system repeatedly encounters under-braced overload, the generative model gradually updates: G is pushed up (you come to expect tougher conditions), and slow baselines such as F^* and ζ^* can shift. Once acute overload is brought under better control, the cumulative D2-biased re-weighting means that ΔV can shrink or even reverse, allowing δ to tilt towards more Creative, exploratory restructuring when current Executive strategies cannot safely contain the load.

- **Risk of overshoot.**

→ If this adjustment overshoots, you risk chronic hypervigilance: high $G - F^*$ even when F is low. In cusp geometry terms, the system may become stuck in a high-engagement Executive basin, with Autopilot and Creative wells effectively collapsed or hard to reach, and the Ψ -band narrows as sustained overload damages meta-states η , χ and λ .

6.4 Dark-Room vs Adventure: $F \ll F^\star$ with different G

In low-load regimes the agent is operating *below* its defended band: actual demand is too low to fully engage its capacities.

- **When $F - F^\star \ll 0$, the system is below the defended band.**

→ It is over-safe and under-stimulated, with a narrow prediction-error landscape and little pressure to change its current policies.

We can distinguish two cases:

Case 1: Low G too ($\xi \ll 0$)

You expect and endorse low stimulation. The system's stance matches the easy reality, so

- ϵ is small; nothing pushes you out.

→ There is no discrepancy between expectation and experience to drive exploration or restructuring.

This yields rigid, risk-averse, over-certain regimes – dark-room tendencies. Autopilot dominates, the cusp flattens around a single low-engagement well, and the Ψ -band is under-used. The agent is *comfortable but stagnant*, with little learning or growth.

Case 2: Higher G ($\xi \geq 0$ while $x \ll 0$)

You expect more complexity than you actually have. Then:

$$\varepsilon = x - \xi \ll 0$$

is strongly negative, signalling *life is too easy compared to what I am geared for*.

That mismatch drives exploration and challenge-seeking:

- raising engagement;
→ the system actively looks for harder problems, richer input or more demanding tasks;
- nudging towards the Creative or exploratory G_f branch;
→ the agent recruits more flexible, generative modes of thinking instead of staying in routine Autopilot;
- pushing F back up into the Ψ -band.
→ actual load increases towards the near-critical corridor where learning, plasticity and high performance are supported.

This is the adventure or hard-training interpretation: increases in free energy can be positively valenced when you are under-critical and treat them as epistemic,

competence-building moves. The valence tag $v(t)$ and σ -map-based valuation together determine whether a given rise in F is coded as good training stress or costly disruption.

7. How the Control System Uses the Interaction

Putting it all together, Trident G describes a control system with a simple outer objective but a richer inner game that uses the interaction between F and G to steer behaviour.

Outer Objective:

$$\text{Minimise } \mathcal{L} = 1/2(F - F^\star)^2$$

That is, to keep actual load near the defended band, not zero. As before, this quadratic is best read as a local approximation to staying on the plateau of high net utility, not as a literal drive to a single exact set-point. In practice, anything within the Ψ -band counts as *good enough*; the loss just penalises excursions that drop the system out of that corridor into chronic overload or underload.

Inner Game:

To realise this outer objective, the inner game is to adjust:

- G (hence $\xi = G - F^\star$), which sets the endorsed challenge level;
- the meta-parameters η , χ and λ , which shape meta-efficacy, coherence and niche coupling;
- the cusp geometry parameters a , b and δ , which determine how many modes exist, how stiff they are and how the system is tilted between exploration and exploitation.

The aim is to make these adjustments so that:

- G tracks typical demands in each niche,
→ expectations become well-calibrated to the environments the agent actually occupies;
- $F^\star$ and the cusp's shape match the agent's competence η , coherence χ and coupling λ ,

→ high-engagement wells are only stabilised when the system is capable of operating there safely and coherently;

- the system spends as much time as possible in the Ψ -band while still learning,
→ it repeatedly revisits near-critical regimes that support high performance and plasticity, without tipping into chronic overload.

In other words, fast dynamics move x around the existing cusp landscape (moment-to-moment shifts in mode and engagement), while slower learning reshapes that landscape so that ‘good’ regions (near-critical, capable and recoverable) are easier to reach, easier to re-enter and easier to dwell in..

Interaction term:

$$\varepsilon = (F - F^\star) - (G - F^\star) = F - G$$

indicates whether the system is:

- **over-prepared** ($\varepsilon < 0$): braced for more load than is actually present;
- **under-prepared** ($\varepsilon > 0$): facing more load than it has braced for;
- **well-calibrated** ($\varepsilon \approx 0$): actual and endorsed load are aligned.

This allostatic prediction error then:

- **drives fast engagement ξ** ,
→ small, transient errors modulate momentary mobilisation (*push a bit harder vs back off*), adjusting ξ within a given niche episode;
- **tilts the cusp via δ (explore versus exploit)**,
→ better-than-expected outcomes bias δ towards exploitative Executive control, whereas worse-than-expected outcomes make exploratory Creative branches more competitive;

- updates slow baselines $G^\star$, $\xi^\star$ and $F^\star$ along with meta-states η , χ and λ on longer time-scales,

→ persistent patterns of over- or under-preparation gradually reshape both the defended band and the agent's beliefs about its own controllability, coherence and coupling to its niche.

In the σ -extended picture, ε also feeds the salience–allostasis hub, which shapes σ -fields and value differences ΔV , further refining how δ responds to different kinds of surprise (threat versus opportunity) by changing which representations and policies are tagged as worth exploring or defending.

Summary of Roles

The short summarising roles are:

- $F - F^\star$: where you actually are relative to the defended band,
- $G - F^\star$: what level of load you expect and endorse,
- ε : the allostatic prediction error that drives learning, engagement shifts and mode switching on the cusp.

In well-learned niches, the system tries to shape things so that x and ξ track each other closely while remaining inside the Ψ -band. When territories change (new jobs, illnesses, social upheavals, environmental shifts), misalignments in this plane are precisely what signal the need for exploration, representational restructuring and re-tuning of meta-states, so that the defended band and the endorsed challenge level can be brought back into register with the new world.

8. Separation of Time-Scales: Meta-parameters vs Physiological Criticality

It is useful to distinguish two layers of change.

8.1 Fast and Semi-Fast: Meta-Parameters and Expectations

Relatively fast-moving quantities include:

- η : meta-efficacy (self-regulatory confidence),
- χ : coherence or meta-uncertainty,
- λ : niche coupling,
- G and $\zeta = G - F^\star$,
- δ : explore–exploit tilt.

These quantities can shift on several intermediate time scales:

- over seconds to minutes via neuromodulators, attention shifts and momentary belief changes;
- over hours to days via standard synaptic plasticity, therapeutic conversations and feedback;
- over weeks to months through training, psychotherapy and lifestyle implementation.

In mammals, a natural neural candidate for the explore–exploit tilt δ is the balance of activity and plasticity between D1-dominant *Go* and D2-dominant *NoGo* cortico-striatal pathways in the basal ganglia. In standard reinforcement-learning accounts, phasic dopamine bursts following better-than-expected outcomes drive long-term potentiation at D1-expressing direct-pathway synapses and long-term depression at D2-expressing indirect-pathway synapses, biasing future choice towards repeating the recently successful policy. This is the canonical exploit move: stabilising an action that has just proved better than the model predicted. Conversely, dopamine dips following worse-than-expected outcomes favour long-term depression in the direct pathway and long-term potentiation in the indirect pathway, weakening the just-selected channel and strengthening *NoGo* weights, thereby discouraging its reuse and making alternative policies more competitive. In Trident G we interpret this D2-weighted shift -

especially when combined with prefrontal D2 effects on cognitive flexibility - as opening the door to Creative, exploratory search over new strategies, whereas D1-dominant patterns support more stable, Executive exploitation of existing ones. On this picture, δ summarises the current D1–D2 bias, shaped by both fast phasic dopamine and the accumulated long-term potentiation and depression history in these loops, which then projects up into large-scale control networks (FPCN-A/B, salience networks) that implement Trident G's mode switches and strategy use.

From this perspective, phasic dopamine bursts and dips are the main carrier of prediction-error information that updates δ and the relevant cortico-striatal synapses on short time-scales, linking the ϵ signal in Trident G to concrete Go/NoGo plasticity. By contrast, slower shifts in tonic dopamine levels modulate baseline excitability and behavioural vigour, and can be viewed as one biophysical contributor to gradual drifts in meta-parameters such as meta-efficacy η and meta-uncertainty χ . We do not assume a strict one-to-one mapping from any single dopaminergic measure to a specific meta-parameter, only that both phasic and tonic modes provide fast and semi-fast levers for the explore–exploit tilt and for how *confident* the system feels operating near the Ψ -band folds.

These meta-parameters effectively reconfigure how the system rides the cusp and occupies the Ψ -band without needing to move the underlying metabolic criticality threshold.

They change:

- how strongly engagement carves distinct Autopilot versus Gf wells;
- how often the system leans into exploration versus exploitation;
- how *safe* it feels operating near folds in the landscape.

Placebo and belief effects are paradigmatic here: they mainly act by changing η , G and interpretations (and, in the σ -extension, the valuation of surprises), rather than the hardware itself.

8.2 Slow: Physiology of Network Criticality

By contrast, the physiological substrate that sets where near-critical dynamics are even possible is slower and more constrained:

- anatomical connectivity and white-matter structure in neural organisms;
- vascular and metabolic capacity, or more generally energy supply and transport;
- baseline excitation–inhibition balance in regulatory circuits;
- endocrine milieu and long-term health.

These determine:

- the location of the near-critical window in terms of energy or load (that is, the feasible range for $F^\star$),
- the intrinsic breadth of the underlying Griffiths phase.

They are capable of change (development, ageing, chronic disease, long-term training), but typically on months-to-years time-scales rather than minutes-to-days. Intensive training or illness can gradually shift where the Ψ -band sits inside the viability set and how wide it can be, but these shifts are slow compared with moment-to-moment changes in η , χ , λ and G .

In cognitively sophisticated organisms it is useful to distinguish, within this slower layer, between (i) very slow changes to basic physiology and criticality thresholds, and (ii) intermediate-rate self-organising changes in the structure of the generative model itself. The latter correspond to what we usually call crystallised intelligence (G_c): semantic, relational and skill structures that emerge from repeated operation near, but slightly below, criticality via

largely unsupervised or self-organised learning processes. These slowly shaped structures provide the substrate on which faster, prediction-error-driven control (Gf-like fluid reasoning and flexible action selection) can operate, allowing the system to reuse and recombine stable patterns rather than relearn them from scratch on each encounter.

8.3 Why This Separation Matters

For most cognitive, training and placebo-like phenomena, it is computationally sensible to treat some parts of the system as effectively fixed over short horizons and let others move quickly.

For modelling purposes, we can:

- **Treat basic criticality thresholds, the underlying Griffiths-phase structure and the slowly self-organising Gc substrate as relatively invariant on day-to-week scales.**

These provide the slowly changing scaffold that determines where near-critical operation is even possible.

- **Let η , χ , λ and G do the quick work of reconfiguring the system's stance on that fixed scaffold.**

These meta-parameters and expectations can change rapidly in response to feedback, therapy or training.

In this picture:

- **Meta-parameters plus fast prediction-error learning in the Gf branch handle rapid adjustments of stance and mode use on the existing cusp landscape.**

They decide, moment by moment, how strongly to engage, whether to explore and which modes to occupy.

- **Slower, largely self-organised learning shapes the Gc branch, gradually refining the structure of the generative model (semantic, relational and skill patterns) on which those fast inferences operate.**

Over repeated exposure, the system builds stable knowledge and skills that can be recombined without relearning from scratch.

- **The very slow physiology of network criticality defines the ultimate limits on where the near-critical window can sit and how broad it can be.**

Development, lifestyle, ageing and chronic disease set the long-term boundaries of the Ψ -band.

So:

- Meta-parameters plus placebo and training are fast-moving knobs that change how the agent uses its existing critical band and Gc substrate.

They alter how boldly the system operates near the fold and how often it revisits near-critical regimes.

- **Physiology of network criticality and self-organising Gc structure are the slower substrates that define where that band sits and how rich and wide it can ultimately be.**

They set the long-run “hardware” limits on viable engagement and cognitive richness.

From a modelling point of view, holding the criticality threshold and Gc substrate roughly fixed while letting η , χ , λ and G adapt quickly is a natural and tractable separation of scales. When needed (for example, developmental or longitudinal work), the slower drift of $F^\star$, the Griffiths zone and the self-organised Gc structure can then be added as higher-level processes.

9. Summary

Framed in Marr's terms, Trident G offers a computational-level story: An organism must maintain itself within a defended, near-critical band of allostatic load while exploiting and learning from its niche. This demands at least:

- $F - F^\star$: a current deviation;
- $G - F^\star$: an endorsed or expected deviation;
- $\varepsilon = F - G$: a miscalibration signal.

These quantities are embedded in a cusp-shaped control geometry over $x = F - F^\star$, with:

- effective engagement:

$$\xi_{\text{eff}} = \lambda \xi = \lambda (G - F^\star)$$

(or more precisely $\lambda(\xi^\star + \tilde{\xi})$);

- δ : an explore–exploit tilt;
- η, χ and λ : meta-parameters slowly reshaping the landscape.

The organism's modes (Autopilot, Executive, Creative and dysregulated) correspond to regions on this cusp surface, with mode switches and learning driven by the evolution of F , G and $F^\star$, the prediction error ε and the meta-states. In the σ -extended version, salience and valuation fields further refine how δ and engagement respond to different kinds of surprise (threat versus opportunity), based on pooled internal (interoceptive) and external (exteroceptive) signals.

On this view, Trident G is a strong candidate for a generic computational description of what any allostatic, learning organism must approximately be doing to live *on the cusp*: staying inside a near-critical Ψ -band where it can both survive and think, using prediction errors and

meta-parameters to shape when to exploit, when to explore and how boldly to operate near the edge of its own viability band.

From this perspective, general intelligence g can be read off the control loop as a long-run statistic over Ψ -band navigation. Let $\text{Occ}_\Psi(t)$ indicate whether the current state $(x, \xi, \delta, \eta, \chi, \lambda)$ lies in the Ψ -band and write $\langle \text{Occ}_\Psi \rangle$ for its time-average under standardised challenges. Let G denote the set of recurrent Ψ -band trajectories (*g-loops*) the organism can realise and reliably re-enter. Then, at the computational level, a natural working definition is that g is an increasing function of (i) $\langle \text{Occ}_\Psi \rangle$ and (ii) the ‘volume’ or diversity of G in $(x, \xi, \delta, \eta, \chi, \lambda)$ -space. Informally, more intelligent organisms spend more time in their Ψ -band and can access a richer repertoire of near-critical loops without falling out of their viability band.

Readings

Free Energy Principle & active inference

Friston, K. (2010). The free energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138. <https://doi.org/10.1038/nrn2787>

Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017). Active inference: A process theory. *Neural Computation*, 29(1), 1–49.
https://doi.org/10.1162/NECO_a_00912

Allostasis, allostatic load & no single set-point

Sterling, P., & Eyer, J. (1988). Allostasis: A new paradigm to explain arousal pathology. In S. Fisher & J. Reason (Eds.), *Handbook of life stress, cognition and health* (pp. 629–649). John Wiley & Sons.

McEwen, B. S., & Stellar, E. (1993). Stress and the individual: Mechanisms leading to disease. *Archives of Internal Medicine*, 153(3), 209–221.
<https://doi.org/10.1001/archinte.153.3.209>

McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *The New England Journal of Medicine*, 338(3), 171–179. <https://doi.org/10.1056/NEJM199801153380307>

Criticality & Griffiths phases in brain networks

Beggs, J. M., & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35), 11167–11177.

<https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003>

Shew, W. L., & Plenz, D. (2013). The functional benefits of criticality in the cortex. *The Neuroscientist*, 19(1), 88–100. <https://doi.org/10.1177/1073858412445487>

Moretti, P., & Muñoz, M. A. (2013). Griffiths phases and the stretching of criticality in brain networks. *Nature Communications*, 4, Article 2521. <https://doi.org/10.1038/ncomms3521>

Metastability & scale-free temporal structure

Linkenkaer-Hansen, K., Nikouline, V. V., Palva, J. M., & Ilmoniemi, R. J. (2001). Long-range temporal correlations and scaling behaviour in human brain oscillations. *Journal of Neuroscience*, 21(4), 1370–1377. <https://doi.org/10.1523/JNEUROSCI.21-04-01370.2001>

Hardstone, R., Poil, S.-S., Schiavone, G., Jansen, R., Nikulin, V. V., Mansvelder, H. D., & Linkenkaer-Hansen, K. (2012). Detrended fluctuation analysis: A scale-free view on neuronal oscillations. *Frontiers in Physiology*, 3, Article 450.

<https://doi.org/10.3389/fphys.2012.00450>

Goldberger, A. L., Amaral, L. A. N., Hausdorff, J. M., Ivanov, P. C., Peng, C.-K., & Stanley, H. E. (2002). Fractal dynamics in physiology: Alterations with disease and ageing.

Proceedings of the National Academy of Sciences, 99(Suppl. 1), 2466–2472.

<https://doi.org/10.1073/pnas.012579499>

Catastrophe theory, cusps & dynamical systems

Thom, R. (2018). *Structural stability and morphogenesis: An outline of a general theory of models* (Advanced Book Classics). CRC Press. <https://doi.org/10.1201/9780429493027>

Zeeman, E. C. (1976). Catastrophe theory. *Scientific American*, 234(4), 65–83.
<https://doi.org/10.1038/scientificamerican0476-65>

Explore–exploit trade-off & neuromodulatory control

Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus–norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403–450. <https://doi.org/10.1146/annurev.neuro.28.061604.135709>

Cohen, J. D., McClure, S. M., & Yu, A. J. (2007). Should I stay or should I go? How the brain weighs the costs of waiting. *Current Opinion in Neurobiology*, 17(2), 210–216.
<https://doi.org/10.1016/j.conb.2007.07.003>

Large-scale brain networks, salience & control

Seeley, W. W., Menon, V., Schatzberg, A. F., Keller, J., Glover, G. H., Kenna, H., Reiss, A. L., & Greicius, M. D. (2007). Dissociable intrinsic connectivity networks for salience processing and executive control. *Journal of Neuroscience*, 27(9), 2349–2356.
<https://doi.org/10.1523/JNEUROSCI.5587-06.2007> (Physiology Journals)

Menon, V. (2011). Large-scale brain networks and psychopathology: A unifying triple network model. *Trends in Cognitive Sciences*, 15(10), 483–506.
<https://doi.org/10.1016/j.tics.2011.08.003>

Neurovisceral integration, autonomic flexibility & HRV

Thayer, J. F., & Lane, R. D. (2000). A model of neurovisceral integration in emotion regulation and dysregulation. *Journal of Affective Disorders*, 61(3), 201–216.

[https://doi.org/10.1016/S0165-0327\(00\)00338-4](https://doi.org/10.1016/S0165-0327(00)00338-4)

(Goldberger et al. 2002 above is also directly relevant here as the core fractal/complexity reference for cardiovascular dynamics.)

Interoceptive inference & constructed emotion (linking (F), (G) and valence)

Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. *Nature Reviews Neuroscience*, 16(7), 419–429. <https://doi.org/10.1038/nrn3950>

Successor representation, predictive maps & cognitive geometry

Stachenfeld, K. L., Botvinick, M. M., & Gershman, S. J. (2017). The hippocampus as a predictive map. *Nature Neuroscience*, 20(11), 1643–1653. <https://doi.org/10.1038/nn.4650>

Gershman, S. J. (2018). The successor representation: Its computational logic and neural substrates. *Journal of Neuroscience*, 38(33), 7193–7200.

<https://doi.org/10.1523/JNEUROSCI.0151-18.2018>

Human intelligence, (g) & cognitive factors

Carroll, J. B. (1993). *Human cognitive abilities: A survey of factor-analytic studies*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511571312> (Google Books)

Deary, I. J. (2012). Intelligence. *Annual Review of Psychology*, 63, 453–482.

<https://doi.org/10.1146/annurev-psych-120710-100353>

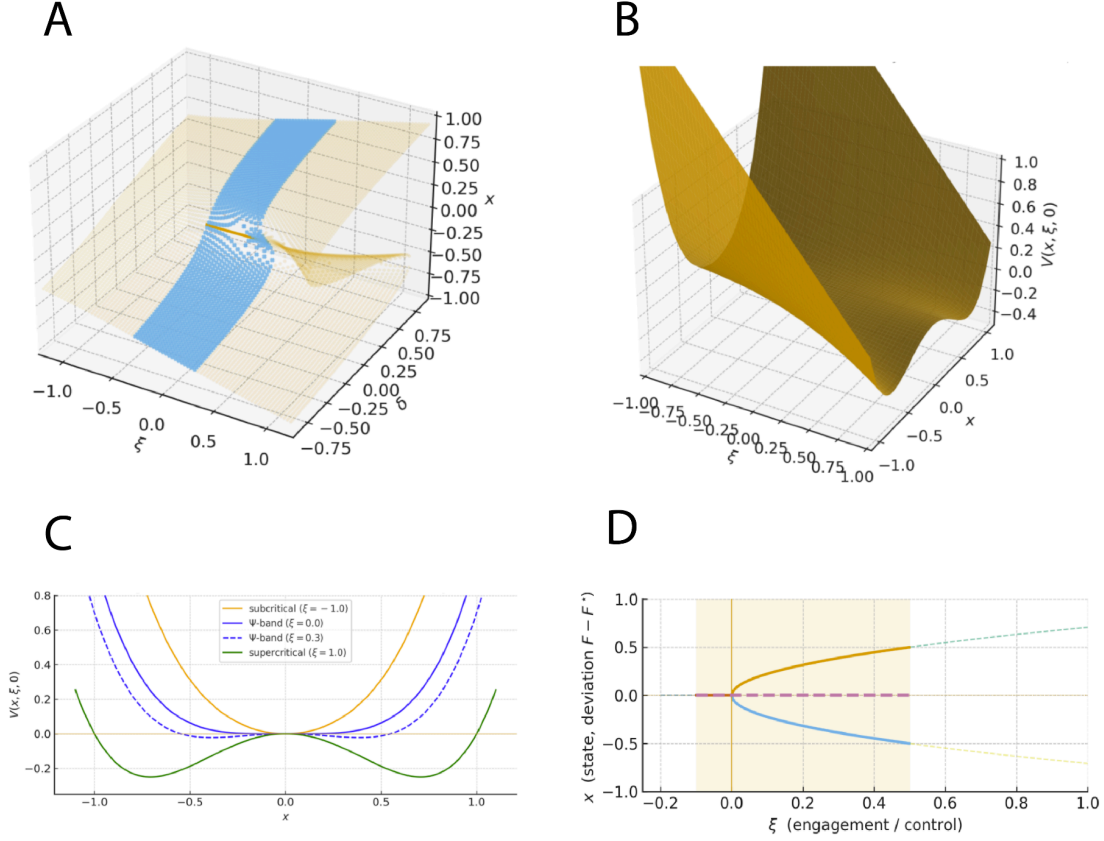


Figure 1. Trident G cusp geometry across regimes.

A. Equilibrium cusp surface in the (ξ, δ, x) space for the normalised potential $V(x; \xi, \delta) = x^4 - \xi x^2 + \delta x$, with meta-states η, χ and λ held fixed so that ξ can be read as a normalised effective engagement parameter (proportional to ξ_{eff}). Each point lies on the stationary set $\partial V / \partial x = 0$. Grey points show the full equilibrium surface; darker points indicate an illustrative Ψ -band collar, here taken as the near-critical region ($|\xi| \leq \varepsilon$) where the potential is shallow and, for $\xi > 0$, the double wells have low barrier heights. The solid curve shows an example Ψ -band g-loop at

zero tilt ($\delta = 0$), running along the central branch near $x \approx 0$ where switching between modes is easy.

B. Normalised cusp potential $V(x; \zeta, 0) = x^4 - \zeta x^2$ at zero tilt, plotted over state deviation $x = F - F^\star$ and engagement-like control parameter ζ (horizontal axes); the vertical axis is the potential $V(x, \zeta, 0)$. For $\zeta < 0$, the surface has a single central basin (low- $|x|$ Autopilot or Gc-like operation). As $\zeta \rightarrow 0$, this basin flattens into a near-critical Ψ -band region where barriers are low and switching is easy. For $\zeta > 0$, the central point becomes unstable and two outer bowls appear at larger $|x|$, corresponding to higher-load, metabolically costly Gf states that the system can enter under high engagement.

C. One-dimensional slices of the normalised cusp potential $V(x; \zeta, 0) = x^4 - \zeta x^2$ for representative values of ζ . The horizontal axis is state deviation $x = F - F^\star$; its magnitude $|x|$ measures deviation from the defended set-point (overall cognitive or metabolic load), while the sign simply labels two symmetric branches. For subcritical engagement ($\zeta = -1.0$) the landscape has a single low- $|x|$ well (Autopilot or Gc-like operation). As ζ approaches and then crosses the cusp ($\zeta = 0.0, 0.3$), the central basin flattens and begins to split, marking the Ψ -band where barriers are low and switching between modes is easy. For supercritical engagement ($\zeta = 1.0$), two deep high- $|x|$ wells appear, representing two qualitatively distinct high-load Gf branches that can be biased towards exploration-dominated versus exploitation-dominated problem-solving when a small tilt $\delta \neq 0$ is present.

D. Trident projection of equilibrium states in the (ζ, x) plane for the normalised cusp potential $V(x; \zeta, \delta) = x^4 - \zeta x^2 + \delta x$ at zero tilt ($\delta = 0$). The vertical axis is state deviation $x = F - F^\star$; $|x|$ indexes overall cognitive or metabolic load relative to the defended set-point. The lightly

shaded band marks an illustrative Ψ -band range of engagement (from slightly negative ζ up to $\zeta = \varepsilon$), capturing the upper end of the Autopilot shaft and the low-barrier Gf regime. For slightly negative engagement the system follows the Autopilot shaft near $x \approx 0$ (low- $|x|$ Gc-like routine control). As ζ crosses zero into the Ψ -band, the equilibrium splits into two high- $|x|$ branches (solid curves), corresponding to two high-engagement Gf modes that, when a small state-dependent tilt $\delta(t)$ is applied, are preferentially used for exploration-dominated versus exploitation-dominated processing. The heavy dashed segment at $x \approx 0$ within the shaded band marks the unstable cusp knife-edge, where small tilts or noise can send processing into one branch or the other. Beyond the Ψ -band ($\zeta > \varepsilon$), the same branches continue as dashed curves, representing more deeply committed high-load Gf states outside the near-critical corridor.

Appendix A. Illustrative Mapping

Table 1 provides a heuristic illustration of how the Trident G quantities could, in principle, be instantiated in non-neural organisms such as bacteria and plants.

In both cases, $F(t)$ should be read as a coarse allostatic load proxy: the instantaneous energetic and regulatory strain required to keep key variables within a viability band. $F(t)$ is the realised load, and $G(t)$ plays the role of anticipated or allostatically braced-for load, implemented via predictive regulatory schemes such as catabolite repression and anticipatory gene networks in microbes, or circadian and systemic defence priming in plants.

The prediction error $\varepsilon(t) = F(t) - G(t)$ then corresponds to a mismatch between realised and expected demand, driving changes in gene expression, growth patterns and defensive investment. Engagement $\zeta(t)$ captures the organism's willingness to incur additional metabolic cost - for example, swimming up a gradient, building a biofilm, diverting carbon into secondary metabolites or committing to flowering.

The near-critical or Ψ -band rows point to evidence that gene-regulatory, metabolic and developmental networks in such systems often operate close to critical points, with power-law-like variability and phase-transition behaviour, which is consistent with Trident G's requirement for an extended near-critical regime that supports flexible reconfiguration.

Phenotypic switching phenomena (e.g. normal $\leftrightarrow$ persister cells, biofilm $\leftrightarrow$ planktonic states) then become natural candidates for cusp-like mode transitions driven by environmental coupling and internal meta-parameters.

Table 1

How Trident G concepts can be instantiated in a bacterium or single cell and in a plant.

	In a bacterium or single cell	In a plant
Viability band	Maintain internal pH, ion gradients, metabolite concentrations and membrane potential within survivable bounds - not too low (starvation, energy collapse), not too high (toxicity, osmotic lysis).	Keep photosynthesis rate, water potential and nutrient status between wilt point and damage from excess light or heat; avoid both hydraulic failure and oxidative or photoinhibition damage.
$F(t)$ - actual load	Instantaneous metabolic or allostatic burden: ATP demand versus supply, oxidative stress, protein misfolding and accumulation of toxic metabolites.	Current water stress, photorespiration, pathogen or herbivore damage and resource scarcity; how hard the plant is working to maintain turgor, repair tissue and run defences.
$G(t)$ - expected load	Anticipatory gene regulation that prepares for likely stresses (e.g. catabolite repression patterns tuned to typical nutrient sequences; induction of stress-response proteins before peak stress).	Circadian and environmental priming of stomata and defence pathways (e.g. stomatal behaviour anticipating daily vapour pressure deficit cycles; systemic acquired resistance primed before likely pathogen attack).
$F^*(t)$ - defended set-point	Preferred internal state balancing growth versus survival (e.g. optimal ribosome and chaperone allocation for a given niche).	Target turgor pressure, chlorophyll content and hormone balance that support near-optimal growth and repair under typical conditions.

Prediction error $\varepsilon(t) = F(t) - G(t)$	Mismatch between actual and anticipated metabolic demand or stress; drives changes in gene expression, chemotactic behaviour and resource allocation.	Difference between realised and expected water or nutrient supply or attack pressure; alters stomatal aperture, growth direction, root–shoot allocation and defence investment.
Engagement $\zeta(t) = G(t) - F^{\star}(t)$	Readiness to expend energy on costly behaviours such as motility, biofilm formation, secretion systems or virulence factors (high ζ) versus energy-saving, low-activity states (low ζ).	Readiness to invest in costly growth and reproduction (elongation, flowering, seed production) versus conservative strategies (root investment, defence, dormancy). High ζ implies aggressive growth or defence; low ζ implies energy conservation.
Near-criticality / Ψ -band	Regimes where gene-regulatory and metabolic networks sit near an order–chaos boundary, with large dynamic range and sensitivity (e.g. critical-like dynamics in genetic networks and metabolic oscillations).	Regimes where transport and signalling networks (phloem transport, stomatal oscillations, electrical and hormonal signalling) show rich variability and long-range correlations, supporting flexible growth and defence responses.
Mode switching	Transitions between planktonic versus biofilm states; growth versus sporulation; motile chemotaxis patterns; normal versus persister phenotypes in fluctuating environments.	Transitions between vegetative and reproductive growth; shade-avoidance versus light-harvesting architectures; shifts between growth-focused and defence-focused regimes; C3 versus

		CAM-like operation under stress in facultative species.
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Appendix B. Socially Extended Trident G

The main text formulates Trident G at the level of individual organisms. However, many real-world forms of intelligence are deeply social and cultural. Appendix B sketches how the same computational quantities can, in principle, be extended to dyads, teams and collectives, using ideas from synchrony, shared generative models and generalised synchrony.

Trident G is formulated for any system that:

- maintains itself within a viability set (a band of allostatic load or free energy compatible with continued existence);
- uses prediction and control to remain in a near-critical Ψ -band where flexible, intelligent behaviour is possible.

In the body of the paper, the system is a single organism. In social settings, agents are often coupled so tightly that it is useful to treat a group (e.g. a dyad, a team, a colony) as a higher-level agent with its own generative model and allostatic load. Under this view:

- individuals remain Trident G agents in their own right,
- but a subset of their states (shared beliefs, norms, roles, joint policies and interaction patterns) form a group-level generative model, with its own quasi-Markov blanket relative to the wider environment.

The same core quantities can then be defined at the group scale:

- $F_{\text{group}}(t)$: group-level allostatic load - overall stress or strain on the collective, including resource scarcity, conflict, coordination cost and risk exposure.
- $G_{\text{group}}(t)$: expected group load - how demanding the group tacitly expects the shared niche to be (e.g. typical market conditions for a firm, predation pressure for a herd, regulatory intensity for a House).

- $F_{\text{group}}^{\star}(t)$: defended group set-point - a reference level of group strain compatible with sustainable operation and near-critical flexibility.
- $G_{\text{group}}^{\star}(t)$: baseline expected group load - the slowly learned sense of *how tough life usually is* for that group in that niche.

From these we can construct:

$$x_{\text{group}}(t) = F_{\text{group}}(t) - F_{\text{group}}^{\star}(t)$$

$$\xi_{\text{group}}(t) = G_{\text{group}}(t) - F_{\text{group}}^{\star}(t)$$

$$\varepsilon_{\text{group}}(t) = F_{\text{group}}(t) - G_{\text{group}}(t) = x_{\text{group}}(t) - \xi_{\text{group}}(t)$$

The group's Ψ -band is then the region in $(F_{\text{group}}, G_{\text{group}})$ space where:

- collective load is neither chronically overwhelming nor chronically under-stimulating,
- the group maintains high dynamic range in its internal coordination (multiple workable coordination modes),
- and recoverability from perturbations (shocks, conflicts and environmental surprises) is high.

The cusp construction can be lifted to this group level by treating x_{group} as the state axis and ξ_{group} and δ_{group} (group-level explore–exploit tilt) as control axes, yielding a group cusp surface whose wells correspond to different coordination regimes (e.g. fragmented versus aligned, exploit versus explore, routine versus creative).

For a social Trident G to be meaningful, individual agents must not merely model each other (Theory of Mind), but become dynamically coupled so that a shared, higher-level structure emerges. Empirically, this is suggested by several forms of synchrony:

Neural and Physiological Synchrony

In humans, interpersonal synchrony in brain activity, heart rate and other autonomic signals is reliably observed during joint action, conversation and cooperation. Such synchrony predicts shared understanding, rapport and performance. From a Trident G perspective, this is what we expect when individuals' internal trajectories are pulled into generalised synchrony by reciprocal prediction and signalling.

Behavioural Synchrony and Joint Rhythms

Walking in step, moving in time, speaking in complementary rhythms and aligning postures all reduce the cost of mutual prediction. These behaviours effectively sharpen the group's cusp (clearer, more stable coordination modes) while lowering individual prediction errors.

Symbolic Communication and Shared Narratives

Language, shared stories, rules and norms allow the group to build and maintain a shared generative model that persists beyond any momentary interaction. This is the cultural substrate of $G\star_{\text{group}}$ and $F\star_{\text{group}}$: long-run expectations about *how we live*, *how we operate* and *what we can handle together*.

In Trident G terms, these mechanisms increase niche coupling at the group level, λ_{group} , and reshape group meta-states such as:

- η_{group} : collective meta-efficacy (belief in the group's controllability and competence);
- χ_{group} : group coherence or shared uncertainty;
- λ_{group} : strength of coupling between group internal models and their shared niche.

As these meta-parameters change, they reshape the group cusp geometry (depth and stiffness of coordination wells, width of the group Ψ -band), in the same way individual meta-states reshape an individual's cusp.

This socially extended picture is not limited to humans:

Human dyads and teams:

Two people solving a problem or performing a joint task tend to converge on shared expectations G_{group} (*this will be hard; we need to push*) and a shared sense of acceptable strain $F\star_{\text{group}}$. Their mutual adjustment of pace, tone, roles and risk-taking can be read as coordinated movement on a dyadic cusp, switching between Autopilot-like routines (well-practised roles) and high-engagement creative or exploit modes.

Bacterial populations and quorum sensing:

Quorum-sensing bacteria collectively adjust gene expression based on pooled chemical signals. The population as a whole behaves as if it had an extended generative model of environmental conditions (*we are many* versus *we are few, the host is in this state*). Biofilm formation, virulence activation and persister formation are then group-level mode switches on a population cusp, driven by shared G_{group} and $\varepsilon_{\text{group}}$.

Swarms, flocks and schools:

Flocking and schooling rely on simple local rules, but yield emergent group-level navigation and risk management. Here, F_{group} could be read as overall exposure to predation and energetic cost of movement; G_{group} as the swarm's implicit expectation of risk and resource availability; and sudden reconfigurations (turns, splits, density changes) as rapid transitions between wells on a group cusp surface.

In each case, the Trident G quantities can be interpreted at the collective scale, provided there is sufficient coupling and shared structure that the group behaves as a semi-coherent agent.

This Appendix is not a full multi-scale formalism, but a sketch of how the same computational story could extend beyond single organisms:

- individual agents still minimise their own $\varepsilon(t) = F(t) - G(t)$, and maintain their own Ψ -bands;
- dense coupling (through synchrony, communication, shared artefacts and environments) makes their trajectories cohere, giving rise to a group-level Trident G with its own $(F_{\text{group}}, G_{\text{group}}, F^{\star}_{\text{group}}, G^{\star}_{\text{group}}, \zeta_{\text{group}}, \varepsilon_{\text{group}})$ and cusp geometry.